

3(a)	force per unit mass	B1
3(b)	changes in height <u>much</u> less than radius of Earth	M1
	so (radial) field lines are almost parallel or $g = GM/R^2 \approx GM/(R + h)^2$	A1

Question	Answer	Marks
3(c)	gravitational force provides/is centripetal force	B1
	$GMm/r^2 = mv^2/r$	C1
	$v = (2\pi \times 1.5 \times 10^{11}) / (3600 \times 24 \times 365) = 2.99 \times 10^4 \text{ (ms}^{-1}\text{)}$	C1
	$6.67 \times 10^{-11} M = 1.5 \times 10^{11} \times (2.99 \times 10^4)^2$	C1
	$M = 2.0 \times 10^{30} \text{ kg}$	A1
	or	
	$GMm/r^2 = mr\omega^2$	(C1)
	$\omega = 2\pi / (3600 \times 24 \times 365) = 1.99 \times 10^{-7} \text{ (rads}^{-1}\text{)}$	(C1)
	$6.67 \times 10^{-11} M = (1.5 \times 10^{11})^3 \times (1.99 \times 10^{-7})^2$	(C1)
	$M = 2.0 \times 10^{30} \text{ kg}$	(A1)
	or	
	$T^2 = 4\pi^2 r^3 / GM$	(C2)
	$M = 4\pi^2 \times (1.5 \times 10^{11})^3 / (\{3600 \times 24 \times 365\}^2 \times 6.67 \times 10^{-11})$	(C1)
	$= 2.0 \times 10^{30} \text{ kg}$	(A1)